

TTTC2343 Project Instruction: Customized Robot Simulation Development

Develop a customized TIAGo robot simulation using ROS and Gazebo. The project includes environment development, autonomous navigation, and robot task execution such as QR code scanning or pick-and-place operations.

Week 7: Draft Proposal

- Requirements:
 - Project title
 - Problem statement
 - Objectives
 - Proposed environment
 - Robot task
 - System flowchart
- Deliverables:
 - Proposal document
 - System architecture

Week 8: Customize Environment in Gazebo

- Requirements:
 - Minimum 3 areas/rooms
 - Obstacles and task stations
 - TIAGo robot integration
- Deliverables:
 - Gazebo world file
 - Screenshots
 - Launch file

Week 9: Mapping and Multi-Point Navigation and UI Development

- Requirements:
 - Generate map
 - Minimum 3 navigation checkpoints
 - Obstacle avoidance
 - User interface (UI) for robot control
 - Receive user input for destination selection

- Navigate autonomously to selected location
 - Display robot status through UI
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- Deliverables:
 - Map files
 - Navigation demo video
 - ROS navigation package

Week 10: Perform Task Action

- Requirements:
 - QR code scanning OR Pick-and-place operation
- Deliverables:
 - Functional ROS package
 - Task demo video

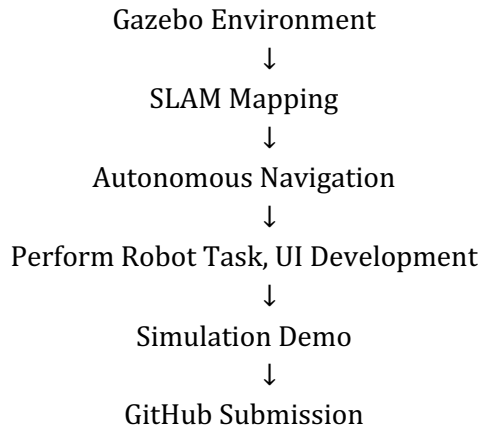
Week 11: Finalize Project and GitHub Upload

- Requirements:
 - Integrate all modules
 - Record full simulation
 - Upload project to GitHub
- Deliverables:
 - GitHub repository link
 - Final simulation video

Final Demo, Report, and Presentation

- Final Report:
 - Introduction
 - Methodology
 - System Design
 - Navigation and Task Execution
 - Results and Discussion
 - Conclusion
- Presentation:
 - 10–15 minute presentation
 - Demo and Q&A session

Suggested Workflow



Evaluation Scheme: Customized TIAGo Simulation Project

Overall Marking Scheme

Component	Marks
Proposal	10
Gazebo Environment Development	15
Mapping, Navigation, Task Execution and UI Development	35
System Integration, Demo & GitHub	15
Final Report	15
Presentation & Demonstration	10
Total	100

Final Report Marking (15 Marks)

Criteria	Marks
Technical Content & Methodology	5
System Design & Implementation	5
Results, Discussion & Report Quality	5

Presentation Marking (10 Marks)

Criteria	Marks
Technical Explanation	4
Simulation Demonstration	4
Presentation Delivery & Q&A	2